

Delay & Dopamine: The RL Addiction Loop

Modeling Addiction through Reinforcement Learning

Nikhil Kapasi

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Abstract

This paper presents a simplified computational approach to modeling addiction using reinforcement learning (RL) algorithms, specifically focusing on the connection between temporal difference error (TDE) and dopamine signaling mechanisms. The algorithms implement and compare two models: a standard Q-learning agent representing normal reward learning, and a modified agent incorporating addiction-specific mechanisms including asymmetric learning rates, tolerance development, and cue conditioning. The results demonstrate how, given the appropriate biological mechanisms, initial rewards for drug-seeking behaviour can transform into compulsive use despite diminishing rewards, mirroring key characteristics of addiction. The comparative analysis reveals significant differences in drug seeking behaviour, reward accumulation, and response to environmental signals between the models. This computational framework provides insight into how normal learning mechanisms can be hijacked in addiction and offers a platform for eventual integration with further neurotransmitter modeling and testing potential interventions.

Introduction

Addiction represents one of the most challenging public health issues of our time, characterized by compulsive drug seeking behaviour despite negative consequences [Koob and Volkow, 2016]. The transition from casual drug use to addiction involves complex neurobiological adaptations that hijack natural reward learning systems [Volkow et al., 2016]. Understanding these mechanisms is crucial for developing effective interventions.

Reinforcement learning (RL) provides a powerful computational framework for modeling how organisms learn from experience to maximize rewards [Sutton and Barto, 1998]. The parallel between RL algorithms and dopamine signaling in the brain makes this approach particularly suitable for studying addiction [Schultz et al., 1997]. Specifically, the temporal difference error (TDE) in RL closely resembles the reward prediction error encoded by dopamine neurons [Glimcher, 2011].

This paper aims to develop a computational model of addiction using RL algorithms that incorporate key

neurobiological mechanisms, specifically those involving dopamine. We implement two models for comparison: a standard Q-learning agent representing normal reward learning, and a modified agent incorporating addiction-specific mechanisms. By comparing these models, we can identify how normal learning processes can be subverted in addiction.

The key contributions of this study include:

- Implementation of a Q-learning algorithm with temporal difference error to model dopamine signaling
- Development of an addiction-prone model incorporating asymmetric learning rates, tolerance, and cue conditioning
- Comparative analysis of normal versus addiction-affected learning
- Demonstration of how initially rewarding behaviour can transform into compulsive use despite negative consequences

Temporal Difference Error and Dopamine Signaling

Temporal Difference Error (TDE) represents a fundamental concept in reinforcement learning that models the difference between expected and actual rewards. In the context of computational models of addiction, TDE provides a framework for understanding how drugs hijack natural reward learning mechanisms in the brain. The TDE is calculated as:

$$\delta = r + \gamma \max_{a'} Q(s', a') - Q(s, a) \quad (1)$$

where δ is the temporal difference error, r is the immediate reward, γ is the discount factor, and $Q(s, a)$ is the value of taking action a in state s .

This computation closely parallels dopamine signaling in the brain. Schultz et al. [1997] demonstrated that dopamine neurons fire when rewards are better than expected (positive prediction error), show no change when rewards match expectations, and decrease their firing

when rewards are worse than expected (negative prediction error). This pattern serves as a teaching signal that adjusts future reward expectations and shapes behaviour. The biological implementation of this mechanism creates a direct bridge between computational reinforcement learning theories and neurobiological processes underlying addiction.

Glimcher [2011] elaborated on this parallel between reinforcement learning and the brain, noting that “If dopamine neurons are an iteratively computed reward prediction error system, then increasing reward on the current trial should increase firing rates”. This dopamine reward prediction error hypothesis provides a neurobiological motivation for reinforcement learning models. This paper further addresses how phasic dopamine release occurs when rewards exceed expectations, while dopamine activity decreases when rewards fall short of expectations, creating a precise learning signal which can be modeled mathematically.

Drugs of abuse directly stimulate dopamine release, creating artificially large positive prediction errors that strengthen associations between drug-taking actions and their contexts. As described in reinforcement learning literature, these artificially triggered prediction errors lead to pathological learning where drug-seeking behaviour is reinforced despite diminishing effects, a hallmark of addiction. The drugs essentially “trick” the brain’s reward prediction system, creating learning signals much stronger than those produced by natural rewards.

In computational models, the TDE mechanism can be modified to reflect addiction-specific processes. For example, asymmetric learning rates for positive versus negative prediction errors can model enhanced sensitivity to rewards and blunted response to punishments observed in addiction. When implemented in algorithms such as Q-learning [Watkins and Dayan, 1992], this asymmetry produces agents bias the models towards negative rewards even given negative consequences, similar to behaviour seen in substance use disorders.

Background and Related Work

Dopamine Reward Prediction Error

Central to the model in this paper is the dopamine reward prediction error hypothesis, which uses the fact that dopamine neurons encode the difference between expected and received rewards [Schultz, 2016]. This temporal difference error serves as a teaching signal that drives learning in both biological and this RL system.

The computational implementation of this mechanism through temporal difference learning algorithms provides a formal framework for understanding how the brain learns from experience [Niv, 2009]. As Glimcher [2011] notes, “If dopamine neurons are an iteratively computed reward prediction error system, then increasing reward on the current trial should increase firing rates.”

Prefrontal Cortex and Inhibitory Control

A critical component of addiction involves dysfunction in the prefrontal cortex (PFC), which is responsible for executive functions including inhibitory control [Goldstein and Volkow, 2011]. The PFC normally acts as a “brake” on impulsive behaviours by evaluating long-term consequences of actions [Dalley et al., 2011]. However, chronic drug exposure impairs PFC function, leading to deficits in inhibitory control that contribute to compulsive drug-seeking [Volkow et al., 2011].

Incentive Sensitization and Opponent Process Theories

Two influential theories of addiction—incentive sensitization theory and opponent process theory—provide complementary frameworks for understanding addiction’s persistence.

The incentive sensitization theory, proposed by Robinson and Berridge, suggests that repeated drug exposure causes neural adaptations that render the brain’s reward system hypersensitive to drugs and drug-associated stimuli [Robinson and Berridge, 2008]. Critically, this theory distinguishes between “wanting” (incentive salience) and “liking” (hedonic impact), proposing that addiction involves pathological wanting even when liking diminishes [Berridge and Robinson, 2016].

Opponent process theory, developed by Solomon and Corbit, describes how initial pleasurable drug effects (a-process) trigger opposing negative states (b-process) that strengthen with repeated use [Solomon and Corbit, 1974]. These opponent processes manifest as tolerance and withdrawal, creating a motivational cycle that drives continued drug use to alleviate negative states [Koob and Volkow, 2010].

Computational Models of Addiction

Previous computational approaches to modeling addiction include the work of Redish [2004], who developed a reinforcement learning model of addiction focusing on the role of dopamine in learning and motivation. Dezfouli et al. [2009] proposed a neurocomputational model for cocaine addiction that incorporated multiple learning systems. Keramati et al. [2017] developed a dual-system theory of addiction from a computational perspective, modeling the interaction between goal-directed and habitual systems.

This approach builds on these foundations while incorporating additional neurobiological mechanisms and directly comparing normal and addiction-affected learning.

Methodology

Reinforcement Learning Framework

This paper implements a Q-learning algorithm, a form of temporal difference learning, to model how the brain updates value representations based on prediction errors. (Code included in folder) The standard Q-learning update rule is:

$$Q(s, a) \leftarrow Q(s, a) + \alpha[r + \gamma \max_{a'} Q(s', a') - Q(s, a)] \quad (2)$$

where $Q(s, a)$ is the value of taking action a in state s , α is the learning rate, r is the reward, γ is the discount factor, and s' is the next state. The term in brackets represents the temporal difference error, which parallels dopamine signaling in the brain.

Standard Q-Learning Agent

The standard agent implements the basic Q-learning algorithm with symmetric learning rates for positive and negative prediction errors. This represents normal reward learning unaffected by addiction mechanisms.

Algorithm 1 Standard Q-Learning Agent

```

1: Initialize  $Q(s, a)$  arbitrarily
2: for each episode do
3:   Initialize state  $s$ 
4:   for each step of episode do
5:     Choose action  $a$  from  $s$  using policy derived
       from  $Q$  (e.g.,  $\epsilon$ -greedy)
6:     Take action  $a$ , observe reward  $r$  and next state
        $s'$ 
7:      $\delta \leftarrow r + \gamma \max_{a'} Q(s', a') - Q(s, a) \triangleright$  TD error
8:      $Q(s, a) \leftarrow Q(s, a) + \alpha \delta$ 
9:      $s \leftarrow s'$ 
10:   end for
11: end for
```

Addiction-Prone Q-Learning Agent

The addiction-prone agent incorporates several modifications to model addiction-specific mechanisms:

1. **Asymmetric learning rates:** Different learning rates for positive (α_{pos}) and negative (α_{neg}) prediction errors, with $\alpha_{pos} > \alpha_{neg}$, modeling the enhanced sensitivity to rewards and reduced sensitivity to punishments observed in addiction.
2. **Tolerance:** Diminishing rewards with repeated drug use, implemented as a multiplicative decay of drug reward magnitude.
3. **Withdrawal:** Negative states when drug is absent, implemented as a negative reward that increases with drug exposure.

4. **Cue conditioning:** Environmental triggers for drug-seeking, implemented as state-specific associations with drug rewards.

Algorithm 2 Addiction-Prone Q-Learning Agent

```

1: Initialize  $Q(s, a)$  arbitrarily
2: Initialize tolerance, withdrawal, and cue strength parameters
3: for each episode do
4:   Initialize state  $s$ 
5:   for each step of episode do
6:     Choose action  $a$  from  $s$  using policy derived
       from  $Q$ 
7:     Take action  $a$ , observe reward  $r$  and next state
        $s'$ 
8:     if  $a$  is drug action then
9:       Apply drug sensitivity modifier to reward
10:      Update tolerance (decrease future drug re-
       wards)
11:      Strengthen cue association for state  $s$ 
12:    else
13:      if in withdrawal state then
14:        Apply withdrawal penalty to reward
15:      end if
16:      if cue strength for state  $s >$  threshold then
17:        Apply cue-triggered craving effect to re-
       ward
18:      end if
19:    end if
20:     $\delta \leftarrow r + \gamma \max_{a'} Q(s', a') - Q(s, a) \triangleright$  TD error
21:    if  $\delta \geq 0$  then
22:       $Q(s, a) \leftarrow Q(s, a) + \alpha_{pos} \delta$ 
23:    else
24:       $Q(s, a) \leftarrow Q(s, a) + \alpha_{neg} \delta$ 
25:    end if
26:     $s \leftarrow s'$ 
27:  end for
28: end for
```

Environment Design

The algorithm operates in a simple environment with multiple states representing different contexts and several actions including a designated drug-seeking action. The environment implements:

- State transitions that model different contexts
- Reward structure with higher initial rewards for drug actions
- Tolerance effects that reduce drug rewards over time
- Withdrawal states that emerge after repeated drug use
- Cue conditioning where states become associated with drug rewards

Implementation

Parameter Settings

The key parameters used in this implementation are:

Parameter	Standard Agent	Addiction Agent
Learning rate (α)	0.1	0.1
Positive learning rate (α_{pos})	-	0.3
Negative learning rate (α_{neg})	-	0.02
Discount factor (γ)	0.95	0.95
Exploration rate (ϵ)	0.1	0.1
Drug reward base	2.0	2.0
Tolerance rate	-	0.03
Withdrawal penalty	-	-4.0
Cue conditioning rate	-	0.1

Table 1: Hyperparameters for Standard and Addiction Agents

Simulation Setup

The simulation runs for 1500 episodes with 100 steps per episode. A standard agent which uses default parameters, and an addiction-prone agent, configured with specific parameters to model addiction-like behaviours interact with their environments in parallel under identical conditions. The environments use a RL framework with 10 states and 4 possible actions, including a drug-seeking action. The simulations track various key metrics while running. `Code included in folder`)

Results

Drug Seeking Behaviour

Figure 1 showcases the significantly higher drug-seeking behaviour in the addiction-prone agent (74% vs 42% in the standard agent) demonstrating how addiction-specific mechanisms can lead to compulsive drug use. This aligns with the incentive sensitization theory, which proposes that addiction involves pathological "wanting" that persists even when "liking" diminishes.

The standard agent, with symmetric learning rates, appropriately reduces drug-seeking when rewards diminish. In contrast, the addiction-prone agent, with asymmetric learning rates favoring positive prediction errors, continues to pursue drugs despite diminishing rewards and accumulating negative consequences.

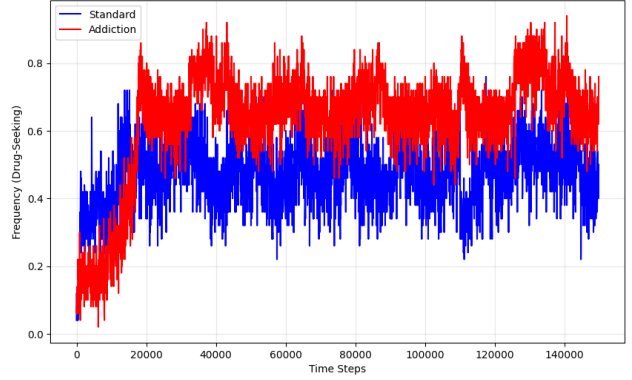


Figure 1: Comparison of drug-seeking behaviour between standard and addiction-prone agents over time. The addiction-prone agent shows significantly higher drug-seeking behaviour.

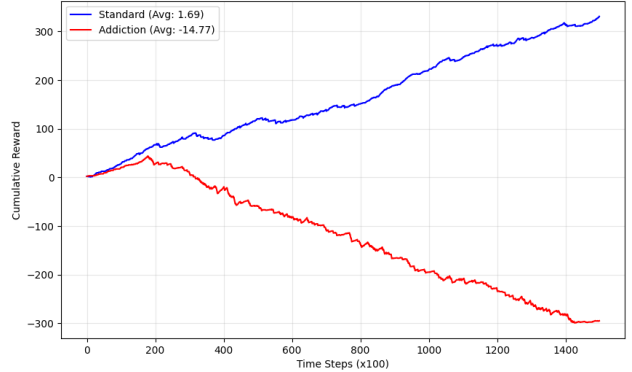


Figure 2: Comparison of total reward accumulation between standard and addiction-prone agents. The standard agent maintains positive rewards (average 1.69), while the addiction-prone agent accumulates increasingly negative rewards (average -14.77).

Reward Accumulation

The divergence in reward accumulation between the two agents (standard: 1.69 average, addiction: -14.77 average) illustrates a key feature of addiction: continued drug use despite negative consequences. This pattern aligns with clinical observations of individuals with substance use disorders who continue drug-seeking despite significant life impairments [Volkow et al., 2016].

Figure 2 clearly shows how the models initially start approximately the same, but the rapid accumulation of negative rewards in the addiction-prone agent reflects the combined effects of tolerance (diminishing drug rewards) and withdrawal (negative states when drug is absent). This creates a vicious cycle where drug use provides less pleasure but becomes increasingly necessary to avoid negative states, consistent with opponent process theory [Koob and Volkow, 2010].

Tolerance Development

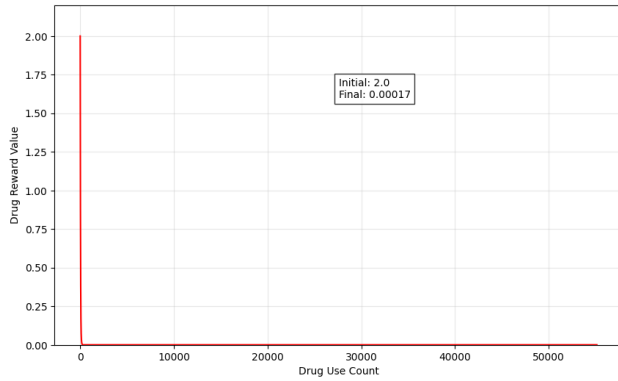


Figure 3: Tolerance development in the addiction-prone agent, showing diminishing drug rewards over time from initial value of 2.0 to 0.00017.

Figure 3 shows the dramatic reduction in drug reward magnitude (from 2.0 to 0.00017) demonstrates the development of tolerance, a hallmark of addiction. This models the neuroadaptations that occur with repeated drug exposure, requiring increasing doses to achieve the same effect [Koob and Volkow, 2016].

Despite this tolerance, the addiction-prone agent continues drug-seeking, illustrating how addiction persists even when drugs no longer produce significant pleasure. This dissociation between reward and behaviour highlights the distinction between the impact of drugs and their motivational significance, a key insight from incentive sensitization theory [Berridge and Robinson, 2016].

Q-Value Evolution

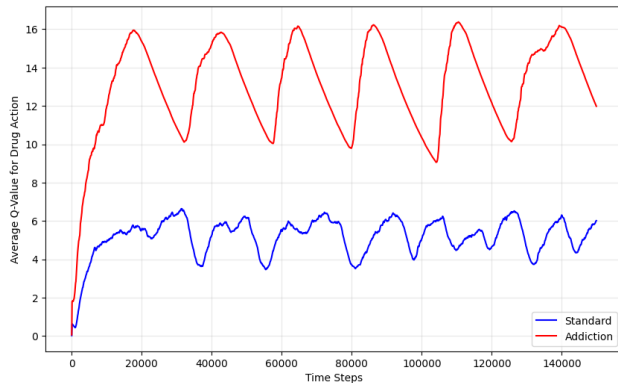


Figure 4: Evolution of Q-values for drug-seeking action in both agents. The addiction-prone agent maintains higher Q-values for drug actions despite diminishing rewards.

Figure 4 showcases the Q value associated with drug seeking behaviour. The maintenance of high Q-values for

drug actions in the addiction-prone agent despite diminishing rewards reveals how addiction can distort value representations. This reflects the asymmetric learning rates, where positive prediction errors have a stronger impact than negative ones, modeling the enhanced sensitivity to rewards and reduced sensitivity to punishments observed in addiction [Frank et al., 2004].

The standard agent’s Q-values more accurately track the actual rewards, leading to more adaptive behaviour. This contrast highlights how addiction can be understood as a dysfunction in the brain’s reward prediction system, where drugs hijack normal learning mechanisms [Redish and Johnson, 2008].

Limitations and Future Work

While the model captures key features of addiction, several limitations should be addressed in future work:

- The current model uses a simplified environment that may not capture the complexity of real-world contexts.
- The implementation focuses on pharmacological aspects of addiction but does not fully address social and environmental factors.
- The model does not incorporate the transition from goal-directed to habitual behaviour that characterizes later stages of addiction [Everitt and Robbins, 2005].

Future work could extend this model by:

- Implementing more complex interactions between parts of the brain and different neurotransmitters
- Implementing a dual-system approach that models both goal-directed and habitual learning [Daw et al., 2005].
- Incorporating social context effects on drug-seeking behaviour [Heilig et al., 2016].
- Modeling individual differences in vulnerability to addiction [Fligel et al., 2011].
- Implementing potential interventions to test their efficacy in reducing drug-seeking behaviour.

Code

All code was written by me and commented for brevity and clarity by ChatGPT (approved by the Professor Pedarsani). The code chunk that was used to produce the figures in this paper and implement both agents is attached in the Zip File. This code chunk is a portion of the code that was used to validate and iterate to the final version of this model that is presented in the code file and this paper.

Conclusion

This paper presents a computational approach to modeling addiction using reinforcement learning algorithms with a focus on temporal difference error and dopamine signaling mechanisms. By comparing a standard Q-learning agent with an addiction-prone agent incorporating asymmetric learning rates, tolerance, withdrawal, and cue conditioning, the model demonstrates how normal learning mechanisms can be hijacked in addiction.

These results show that the addiction-prone agent develops compulsive drug-seeking behaviour despite diminishing rewards and accumulating negative consequences, mirroring key features of addiction. The model provides insights into how addiction can be understood as a dysfunction in the brain’s reward prediction system, where drugs exploit natural learning processes.

This computational framework offers a platform for testing hypotheses about addiction mechanisms and potential interventions. By formalizing theories of addiction in computational terms, we contribute to a deeper understanding of the neurobiological basis of addiction and open avenues for developing more effective treatments.

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